Alireza Safarzadeh

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Professional Summary

Machine Learning Engineer / Data Scientist with 5+ years of experience in fintech analytics, transaction monitoring, and predictive modeling. Skilled in Python (pandas, NumPy, scikit-learn), TensorFlow/Keras, SQL, and end-to-end ML pipelines. Built production models for ATM status prediction, customer churn, and medical-image segmentation. Experienced in reliability analytics on banking switch and core systems, ETL automation, and KPI reporting (availability, MTTR, MTTF).

Core Skills

- **Programming:** Python, SQL, MATLAB
- **ML:** Ensemble learning (RF, GBM, LightGBM, CatBoost), SHAP, cross-validation, feature engineering, class imbalance (SMOTE/ADASYN)
- **Deep Learning:** TensorFlow/Keras, UNet/UNet++, CNN segmentation, mixed precision, EMA, TTA
- **Data Engineering:** ETL pipelines, log processing, Oracle/SQL Server views, batch automation
- **Visualization:** Matplotlib, Seaborn, Plotly
- **Domains:** Fintech (transaction routing, switch/core analytics, ATM reliability), medical imaging

Industry Experience

Pulseware Co. | https://www.pulseware.ir/ — Tehran, Iran

Senior Data Analyst — *Mar 2023 – Present*
Junior Data Analyst — *Feb 2020 – Mar 2023*
Data Analyst Intern — *Nov 2019 – Feb 2020*

Projects

Switch & Core Banking Reliability Analytics — *2019 – Present*

- Analyzed operational logs from the bank's internal switch and core banking system to measure uptime, downtime, outage frequency, and failure patterns over monthly and yearly periods.
- Computed reliability KPIs—including availability, MTTR, MTTF, and failure counts—for both switch and core components.
- Correlated switch-level failures with core-banking responses to identify whether outages originated from the switch or the core platform.
- Developed automated scripts to detect downtime intervals, quantify transaction impact, and categorize failures by component.
- Delivered SLA-focused reliability reports for governance teams and IT operations.

e-Government API Performance & Reliability Analytics — *2020 – Present*

- Analyzed high-volume API call logs for national verification services including Postal Address Verification, National Civil Registry (ID validation), and Shahkar mobile–ID matching.
- Measured availability, timeout frequency, error rates, and latency across banking channels (branches, ATM, mobile banking).
- Generated monthly SLA/compliance reports to evaluate provider performance and detect degradation patterns.

ATM Status Error Reduction with ML (with dashboard integration) — *Nov 2019 – Present*

- Parsed ATM journal text files at scale; engineered features (e.g., exponential interval features) to improve status detection.
- Evaluated multiple models (Decision Tree, Bagging, LightGBM, CatBoost, DES/DCS, Random Forest); selected robust ensembles for deployment.
- Integrated model outputs into the bank's operational dashboard to reduce false alarms and missed outages.

Automated ATM Network Quality Reporting — *Sep 2021 – Present*

- Designed ETL pipelines to extract data from SQL/Oracle views aligned to reporting needs.

- Automated monthly KPI computation (availability, MTTR, MTTF) using MATLAB and integrated the workflow into internal reporting.

Peak Transaction Forecasting (Nowruz Period) *Jan 2024 – Mar 2024*

- Analyzed historical March transaction patterns; engineered temporal/seasonal features and evaluated multiple models.
- Selected Cubic SVM for peak forecasting based on validation performance.

Interactive RTGS Transaction Analyzer (MATLAB UI) *Jan 2022 – Mar 2022*

- Developed an interactive MATLAB UI to select banks and visualize RTGS volumes using dynamic race charts.

Education

University of Tehran *Sep 2018 – Feb 2022*

M.Sc. in Electrical Engineering (Control) *Tehran, Iran*

- Thesis: *Optimizing False Alarms in Automated Teller Machines (ATMs) with Data Fusion Methods*
- Industry Partner: Pulseware Co.

Zanjan University *Sep 2013 – Sep 2018*

B.Sc. in Electrical Engineering (Control) *Zanjan, Iran*

Projects

Brain Tumor Segmentation with Multi-Model Fusion *2024 – Present*

- Implemented a semantic segmentation pipeline on the Brain Tumor MRI dataset using UNet, UNet++, DeepLab, and MobileNet.
- Developed a UNet-based fusion refiner combining multi-model probability maps, achieving Dice = 0.852 and IoU = 0.743.
- Applied Albumentations, mixed precision training, EMA, TTA, BCE+Dice loss, and post-processing (connected component cleanup).

ISEPS: Customer Churn Prediction for Financial Institutions *2024 – Present*

- Built a stacked ensemble (RF, GBM, LightGBM, CatBoost) on a real 10,000-customer dataset.
- Engineered RFM, Balance per Product, Engagement Score, anomaly scores; handled imbalance with ADASYN.
- Improved F1-score from 80% (best single model) to 95% with SHAP-based interpretability.

Optimizing False Alarms in ATMs (M.Sc. Thesis) *2019 – 2022*

- Industrial project improving ATM status detection using ensembles and log-derived features.

Selected Publication

A. Safarzadeh, M. J. Heidarian, R. Sharifanfar, S. F. Haqiqi, and M. Reza Jamali, *"Estimating the Trend of COVID-19 in Iran Before and After the Start of Vaccination,"* 2021 IEEE CSCI, Las Vegas, USA, pp. 1209–1216. DOI link

Teaching

Teaching Assistant, University of Tehran

- Linear Control Systems Lab Sep 2019 – Feb 2020

Interests

Film, Badminton, Setar, Running.